

Speech Enhancement System using Digital Signal Processing

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Abstract—Speech signals are often corrupted by noise during recording and transmission, which reduces their clarity and quality. This paper presents a simple speech enhancement system using digital signal processing techniques. The approach involves adding noise to a clean speech signal and then applying digital filters to reduce the noise. Finite Impulse Response (FIR) and Infinite Impulse Response (IIR) filters are designed and implemented in MATLAB for this purpose. The performance of the filters is evaluated by comparing the original, noisy, and filtered signals using graphical analysis. The results demonstrate that both FIR and IIR filters effectively improve the quality of the speech signal, with each method having its own advantages.

Index Terms—Speech Enhancement, Digital Signal Processing, FIR Filter, IIR Filter, MATLAB

I. INTRODUCTION

Speech is one of the most common ways in which people communicate. In practical situations, speech signals are often affected by unwanted noise during recording or transmission. This noise reduces clarity and makes the signal difficult to understand.

Speech enhancement is the process of improving the quality of a speech signal by reducing noise. It is widely used in applications such as mobile communication, hearing aids, and voice-controlled systems. Various techniques are available for speech enhancement, but many of them are complex and require high computational resources.

In this work, a simple approach is followed using digital signal processing techniques. The focus is on using Finite Impulse Response (FIR) and Infinite Impulse Response (IIR) filters to reduce noise from a speech signal. A clean speech signal is first taken and noise is added to it. Then, FIR and IIR filters are applied to improve the signal quality.

The entire system is implemented in MATLAB. The performance of the filters is studied by comparing the original, noisy, and filtered signals using graphical analysis. This approach helps in understanding the basic concepts of speech enhancement in a simple and practical way.

II. LITERATURE SURVEY

Several methods have been developed for improving the quality of speech signals affected by noise. One of the basic approaches involves the use of digital filters such as Finite Impulse Response (FIR) and Infinite Impulse Response (IIR)

filters. These filters are widely used due to their simplicity and ease of implementation.

FIR filters are known for their stability and linear phase characteristics, which help in preserving the shape of the signal. IIR filters, on the other hand, are more efficient and require fewer coefficients to achieve similar performance.

More advanced techniques such as spectral subtraction and adaptive filtering are also used for speech enhancement. However, these methods are more complex and require higher computational effort.

In this work, simple FIR and IIR filtering techniques are used to enhance speech signals, as they are suitable for basic understanding and practical implementation.

III. METHODOLOGY

The proposed speech enhancement system follows a simple and structured approach using digital filters. The overall process starts with a clean speech signal, which is then affected by noise. The noisy signal is processed using FIR and IIR filters to reduce the noise and improve clarity.

The main steps involved in the system are as follows:

- 1) Load a clean speech signal in MATLAB.
- 2) Add noise to the speech signal to simulate real-world conditions.
- 3) Design an FIR filter using standard filter design functions.
- 4) Design an IIR filter using the Butterworth method.
- 5) Apply both filters to the noisy signal.
- 6) Compare the original, noisy, and filtered signals.

The FIR filter is based on a non-recursive structure and uses only present and past input values. The general form of the FIR filter is given by:

$$y[n] = \sum_{k=0}^N b_k x[n-k] \quad (1)$$

The IIR filter uses a recursive structure, where past output values are also used along with input values. This makes the filter more efficient but slightly complex.

A block diagram of the proposed system is shown in Fig. 1.

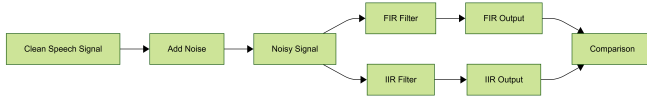


Fig. 1. Block diagram of the speech enhancement system

IV. IMPLEMENTATION

The proposed speech enhancement system is implemented using MATLAB. The process involves loading a speech signal, adding noise, designing filters, and applying them to improve the signal quality.

A. Loading the Speech Signal

A speech signal in .wav format is loaded into MATLAB using the `audioread` function.

```
[x, Fs] = audioread('speech.wav');
x = mean(x, 2);
```

B. Adding Noise

Noise is added to the clean speech signal to simulate real-world conditions.

```
noisy_signal = x + 0.02*randn(size(x));
```

C. Design of FIR Filter

An FIR filter is designed using the `fir1` function.

```
N = 20;
Fc = 0.3;
b = fir1(N, Fc);
fir_output = filter(b, 1, noisy_signal);
```

D. Design of IIR Filter

An IIR Butterworth filter is designed using the `butter` function.

```
order = 4;
Fc = 0.3;
[b, a] = butter(order, Fc);
iir_output = filter(b, a, noisy_signal);
```

E. Plotting the Signals

The original, noisy, and filtered signals are plotted for comparison.

```
t = (0:length(x)-1)/Fs;
subplot(3,1,1); plot(t, x); title('Original Signal');
subplot(3,1,2); plot(t, noisy_signal); title('Noisy Signal');
subplot(3,1,3); plot(t, fir_output); title('FIR Output');
```

V. RESULTS

The performance of the speech enhancement system is evaluated by comparing the original, noisy, and filtered speech signals. The comparison is carried out using time-domain plots.

A. Original and Noisy Signals

The original speech signal is shown in Fig. 2. The waveform is smooth and represents clean speech. After adding noise, the signal becomes irregular as shown in Fig. 3. The fluctuations in amplitude indicate the presence of noise.

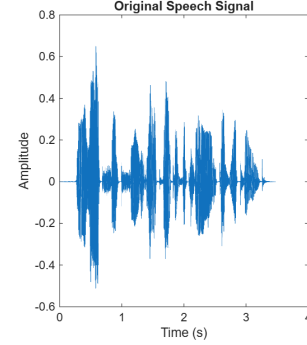


Fig. 2. Original speech signal

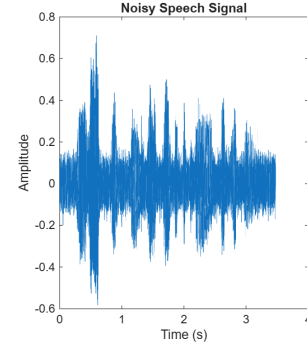


Fig. 3. Noisy speech signal

B. Filtered Signals

The output of the FIR filter is shown in Fig. 4. The noise level is reduced and the signal appears smoother compared to the noisy signal.

The output of the IIR filter is shown in Fig. 5. The signal becomes clearer and the noise is reduced effectively with fewer computations.

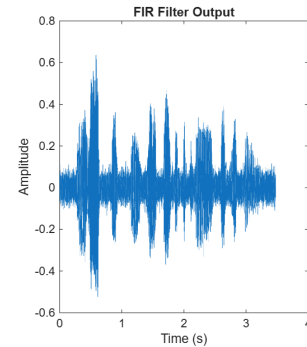


Fig. 4. FIR filter output

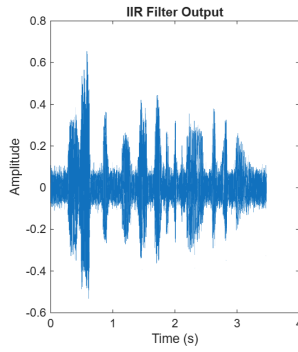


Fig. 5. IIR filter output

From the results, it can be observed that both filters improve the quality of the speech signal. The filtered signals are closer to the original signal compared to the noisy signal.

VI. SUMMARY OF RESULTS

A comparison between the FIR and IIR filters is presented in Table I. The comparison highlights the key characteristics observed during the implementation and analysis.

TABLE I
COMPARISON BETWEEN FIR AND IIR FILTERS

Parameter	FIR Filter	IIR Filter
Structure	Non-recursive	Recursive
Stability	Always stable	Conditionally stable
Phase Response	Linear phase	Non-linear phase
Computation	More computations	Less computations
Efficiency	Lower	Higher
Output Quality	Smooth signal	Clear signal with fast response

From the comparison, it is observed that both filters are effective in reducing noise from the speech signal. The FIR filter provides a stable and smooth output, while the IIR filter achieves faster response with lower computational effort.

VII. CONCLUSION

In this work, a simple speech enhancement system using digital signal processing techniques has been presented. The system focuses on reducing noise from a speech signal using FIR and IIR filters. A clean speech signal was taken, noise was added, and both filters were applied to improve the signal quality.

The results show that both FIR and IIR filters are effective in reducing noise. The FIR filter provides a stable output and maintains the shape of the signal, while the IIR filter achieves similar results with lower computational effort. This makes the IIR filter more efficient for practical applications.

The implementation in MATLAB helps in understanding the basic concepts of digital filtering and speech enhancement in a clear and practical manner. Overall, the work demonstrates that simple filtering techniques can be used to improve speech quality without using complex methods.

VIII. FUTURE WORK

The current work focuses on basic speech enhancement using FIR and IIR filters. In future, the system can be extended by implementing real-time processing for live audio signals. More advanced techniques such as adaptive filtering and spectral methods can also be explored to achieve better noise reduction.

Further improvements can include optimizing filter parameters for different types of noise and testing the system with a wider range of speech samples. The use of machine learning based methods can also be studied for enhanced performance in complex environments.

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